

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

(A State University under Govt. of NCT of Delhi • Established under Delhi Act 06 of 2018)

DEPARTMENT OF INFORMATION TECHNOLOGY



LAB PRACTICAL RECORD FILE PATTERN ANALYSIS & RECOMMENDER SYSTEMS (PARS)

SUBMITTED BY:

Student Name: Monis
Roll Number: 2023UIT3027
Batch & Group: Batch 1 / Group 1
Course / Branch: B.Tech Information Technology

SUBMITTED TO:

Lab Instructor: Dr. Vishal Maheshkar
Department: Department of Information Technology
Institution: Netaji Subhas University of Technology
Campus: Dwarka, Sector-3, New Delhi

Academic Session: 2026–2027

INDEX OF PRACTICAL EXPERIMENTS

Course: Pattern Analysis & Recommender Systems (PARS) Lab | Student: Monis (2023UIT3027)

EXP. #	EXPERIMENT AIM / PROBLEM STATEMENT	PAGE	TEACHER'S SIGN.
01	Prepare a document by studying any two datasets of different domain, write the description of dataset consisting of purpose, what is source, write Feature description, how many features, work done by various authors and performance achieved by atleast three authors in recent years. Cite References and add references.	01–06	
02	Prepare a page using flask consisting of a code in python to extract and store the two datasets that you studied in Experiment 1, Plot the performance plot, display different features having description	07–11	
03	Prepare an informative page using flask consisting of a code in python to extract features from text/video/audio. Extract at least 3 features from various timestamp (at least 10 Samples) justifying the objective for application. text/video/audio file should exist in the page.	12–15	
04	Prepare an interactive page using flask consisting of a code in python for Single feature multiple class, Multiple features multiple class and then classify test samples using Bayesian approach.	16–19	
05	Prepare a page using flask consisting of a code in python to extract text paragraph, separate the sentences, remove various stop words, perform Lemmatization and stemming then classify test samples using Bayesian approach.	20–23	
06	Prepare a page using flask consisting of a code in python to perform Bayesian Risk Minimization for at least three classes, three actions. Given loss function, particular action should trigger particular action, that action should reflect in page itself.	24–27	
07	Implement Linear regression in python (no inbuilt function should be used) to find optimized parameters using Gradient descent for any application and Plot Gradient Descent curve.	28–31	
08	Implement Logistic regression for binary class and multiclass classification in python (no inbuilt function should be used) to find optimized parameters using Gradient descent for any application and Plot Gradient Descent curve.	32–35	
09	Implement regularization for any application show plots of 1)decision boundary and 2) performance before and after applying regularization	36–39	
10	Implement Principal component analysis (PCA) for any application which involves testing. Also show the projected line and new reduced feature mapped on projected line.	40–43	
11	Implement Linear Discriminant analysis (LDA) for any application which involves testing. Also show the projected line and new reduced feature mapped on projected line.	44–47	

Complete Practical Source Code & Interactive Applications: <https://github.com/dSAxmonis/pars-lab>